Second-life Electric Vehicle Batteries 2025-2035: Markets, Forecasts, Players, and Technologies

10-year forecasts: second-life EV batteries & EV battery availability. Analyses: second-life battery technologies, markets, repurposer & end-of-life diagnostics players, regulations, repurposing costs & techno-economic analysis vs first-life Li-ion BESS.

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Report overview

Slides:	296
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The second-life EV battery market is forecast to reach US$4.2B by 2035.

Second-life Electric Vehicle Batteries 2025-2035: Markets, Forecasts, Players, and Technologies

This report provides the following information:

- In-depth analysis on the second-life EV battery market, including funding by repurposers, key player activity, key automotive OEM activity, regional analysis and historic data of second-life batteries in Europe, the US, Africa, Japan, and Australia, and repurposer market share. Further granular analysis includes second-life batteries deployed (<2022-2024) by type of player (repurposer vs automotive OEM) and by depth of disassembly (pack-level, module-level, cell-level).
- Discussion and analysis of business models of second-life battery repurposers, their revenue generation mechanisms, in residential, commercial and industrial (C&I), and grid-scale battery storage markets. Key overview of second-life BESS supply/value chain and applications in key markets.
- Raw data tables for second-life batteries deployed by key repurposer and other key projects by automotive OEMs over time are provided.
- Key conclusions for the second-life EV battery market, including challenges, drivers and opportunities for growth. This includes trends on technology cost, policy, battery design and performance, repurposing costs and automation, advanced battery testing and grading technologies, B2B marketplaces, business models and revenue sharing.

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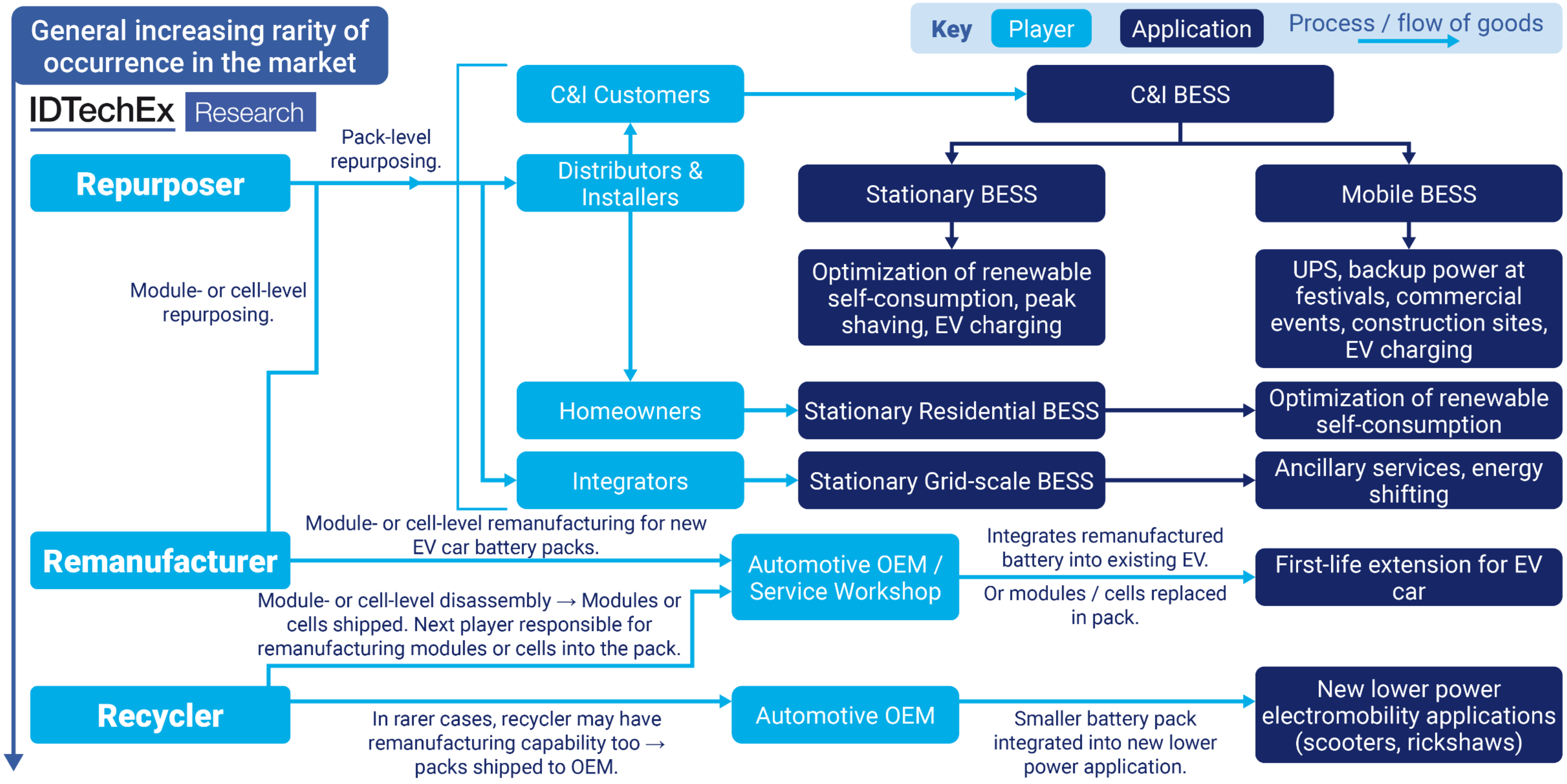
Second-life battery applications and supply chain overview



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Why and when do batteries fail?

Batteries reach End of Life (EOL) for several reasons, and will have undergone different rates of degradation due to different operating conditions. Batteries degrade due to both calendar aging during idle periods as well as cycling aging during operating periods in which they are charged and discharged. As a result, battery capacity is lost over time. Electric car batteries are expected to function for 11-13 years, given the most recent data which suggests this. When the batteries cannot satisfy the requirements for use in electric cars due to capacity loss, for example when the loss of battery capacity limits the driving range of the electric car, batteries are retired from electric vehicles and either a replacement battery is needed or the entire electric car is scrapped.

Some batteries will have degraded faster than others, even of the same battery model, and this will depend on temperature, C-rates, average state of charge (SOCm), and depth of discharge (DOD). For instance, batteries that have operated at temperatures outside of operating standards for too long will have degraded faster.

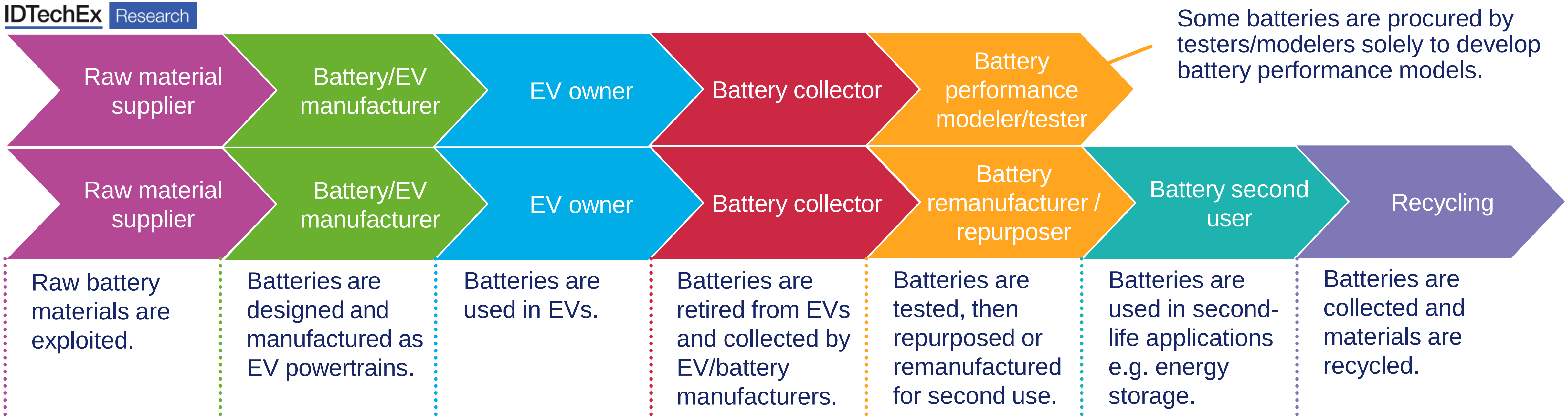
Electrolyte is metastable during battery charging. The formation of the solid-electrolyte interface (SEI) on electrodes prevents further decomposition of solvent, but over time it causes the loss of cyclable lithium ions, and thus capacity fade, as the SEI layer thickens.

During battery charging/discharging, lithium ions move between the two electrodes, making the electrodes shrink and swell. The increasing number of cycles causes particle expansion and structure degradation of active materials, which leads to mechanical fracture and capacity fade of the battery.

At the end of their service life in an electric car, the retired batteries could still retain 70-80% of their initial capacity. In other cases, for example car accidents and pilot fleets, the retired batteries might have more than 80% capacity left.

When batteries are removed from electric vehicles, there are several options to consider: first-life extension in a different EV, remanufacturing/repurposing, and recycling.

Battery second use connects the electric vehicle and battery recycling value chains



The value chain of B2U extends from the first life of the batteries as EV powertrains throughout to the very end of the battery life when they are finally recycled.

Unlike new batteries for stationary storage, B2U connects the value chains of EVs and battery recycling and battery value depends on how they are designed, manufactured and used in their first life in EVs and how they are collected, repurposed and used in their second-life applications.

B2U delays the recycling process and the raw materials feeding back to the start of the value chain.

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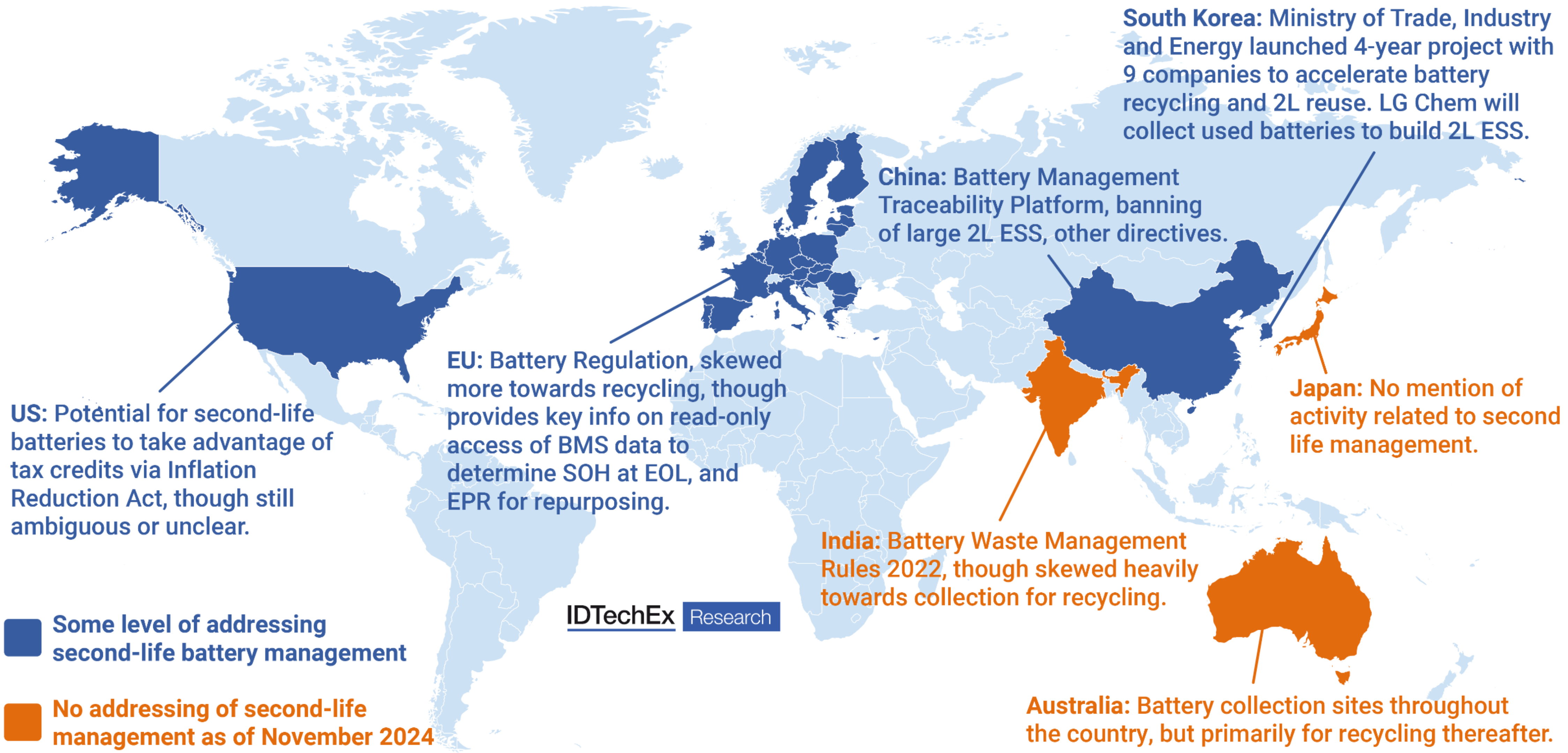
Global second-life EV battery regulatory landscape



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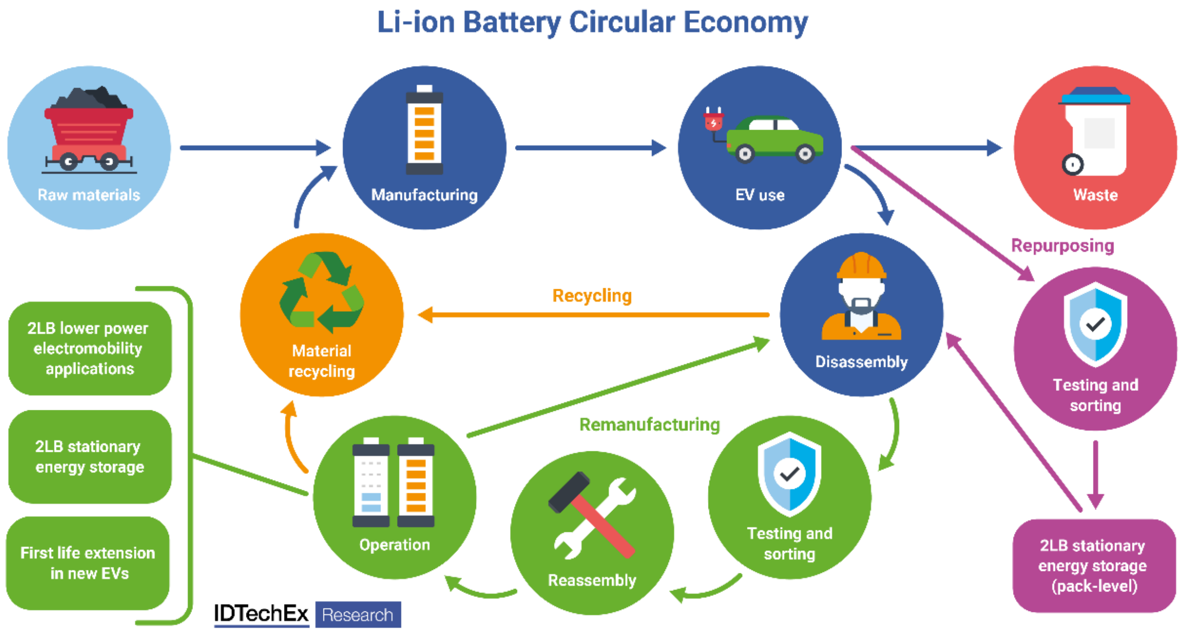
Introduction to the repurposing or remanufacturing process

The Li-ion EV battery circular economy comprises multiple important steps, requiring decisions from key stakeholders at any one part. Second life batteries created through a repurposing or remanufacturing process offer benefits of maximizing battery value and extending battery life.

Remanufacturing and repurposing, generally, are terms that can be used interchangeably, though repurposing is used more-so to describe the repurposing of EV battery packs or modules for second-life BESS, and remanufacturing for replacing some new cells or modules into a pack for battery first-life extension in an EV. The typical steps in the repurposing process include manual inspection for mechanical integrity (cracks, electrolyte leaks), testing / grading, disassembly to pack-, module-, or cell-level, (reassembly of modules into packs if needed), and integration of packs into a container or cabinet to produce second-life BESS. These processes all require some level of manual intervention.

Repurposers could either grade entire packs, modules, or cells, and perform these tests before or after certain disassembly steps. Generally, most repurposers will cycle battery packs to get the overall pack State-of-Health and repurpose at pack-level. Some players choose to do this to bypass more time-consuming and complex disassembly procedures, which will require the use of a skilled workforce to achieve. This also has added benefits of reduced safety risk.

This chapter provides a case study from the literature on disassembling a retired EV battery, as well as cost analysis of the overall repurposing process, from shipment of retired EV batteries through to final containerized second-life BESS production. This has been built up through IDTechEx's primary research with key repurposers in the second-life EV battery market.



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Battery and testing definitions

Battery capacity: The maximum amount of energy a battery contains at a particular point in time.

State of Health (SOH): The ratio of maximum releasable capacity of a battery, under nominal cycling conditions, to its rated capacity.

Electrochemical impedance: The sum of internal resistance (IR) and reactance of a battery. IR is made up of ionic and electric resistance.

Pulse (dis)charging: A method used to measure the consistency in voltage at high C-rates among various batteries being tested.

Cycle testing: A method used to determine the remaining full charge-discharge cycles in a battery before its nominal capacity falls below x% of its initial rated capacity.

State of Charge (SOC): The capacity of a battery that is currently available divided by its rated, nominal capacity.

State of Power (SOP): The maximum power that can be released or absorbed steadily by the battery within a predetermined time interval.

Self-Discharge: A phenomenon in batteries where internal electrochemical processes causes batteries to discharge, irrespective of any connection between the electrodes and any external circuit.

SEI Formation: A component of Li-ion batteries, formed from the decomposition materials associated with the electrolyte of the battery. This is formed on the surface of the anode in a battery cell, and can cause cell internal resistance to increase.

Remaining Useful Life (RUL): An estimation for the time left before a battery first falls below a failure threshold, e.g., in first life batteries, this may be when capacity reaches 80% of its rated capacity. There is no strict definition for RUL for second life batteries.

SEI formation and growth

The solid electrolyte interphase (SEI) layer is a component of Li-ion batteries, formed from the decomposition materials associated with the electrolyte of the battery. This is formed on the surface of the anode in a battery cell, and can cause cell internal resistance to increase.

During the first charge and discharge of a Li-ion battery, the electrode material reacts with the electrolyte at the solid-liquid (S-L) phase interface. Subsequently, a thin film forms on the surface of the electrode material, where Li+ ions can be embedded and removed freely, while electrons cannot. The SEI layer is ~100-120 nm thick, and is mainly composed of various inorganic components.

When a Li-ion battery starts to charge/discharge, Li+ ions are extracted from the active material of the positive electrode, then enter the electrolyte, penetrate the separator, enter the electrolyte and finally integrate themselves between the layered gaps of the anode. Electrons leave the cathode along the outer end loop, and enter the anode. Here, a redox reaction occurs between the electrons, solvent in the electrolyte and Li+ ions. As the thickness of the SEI increases to the point where electrons are unable to penetrate it, a passivation layer forms, inhibiting the redox reaction from continuing. This ultimately reduces the battery ability to perform charging/discharging efficiently.

Parts of the Li+ ions are consumed on forming of the initial SEI film, increasing irreversible capacity of batteries, and decreasing the charge/discharge efficiency of the electrode material. SEI is insoluble in organic solvents, and can exist in stable condition in organic electrolyte solutions.

Therefore, battery testers looking to repurpose retired EV batteries for second life reuse may be interested in understanding the thickness of the SEI layer in these batteries. Those with very thick SEI layers would be rather inefficient for charging/discharging and not have much remaining useful life. Thus, it may be more appropriate to consider using these batteries for lower-energy intensity second life applications or recycling them. However, it would not be practical to perform the tests to determine SEI layer formation and structure, detailed in the following two slides. This is because cell teardowns would be required to gain the anode material to perform these tests, and thus rendering the torn-down batteries useless for remanufacturing for second life applications. IDTechEx postulates that such tests could be performed to help facilitate the build of a more accurate battery performance model. The use of known SEI layer data of a few batteries could represent SEI layer formation in untested batteries (which also originate from the same batch of procured batteries which have been torn down for testing).

Flowcharts for determining RUL

Direct approach (with BMS data)

Extract relevant data from BMS, e.g., current, voltage, temperature, SOH.

↓

Estimate SOH from BMS data (if available) and/or cycle battery for several cycles, collect capacity data

↓

Battery capacity degradation curve

↓

Predict RUL based on SOH data and degradation curve

Note: this approach ages the battery in the process, and would need to be performed for each battery to determine its RUL. Therefore, this is not the most feasible option on a large-scale.

Data-driven approach

Data collection / parameter extraction; temperature-, voltage-, current-, capacity-, related, etc. These features can be extracted from BMS).

↓

Train the algorithm

↓

Predict RUL

↓

Cycle battery to failure threshold

↓

Compare real data to prediction to check accuracy

Physics-based approach

Data collection; requires knowledge of up to 40 battery parameters – from BMS & OEM.

↓

Create physics-based model, based on battery physical / electrochemical processes

↓

Predict RUL by inputting parameters into model

↓

Cycle battery to failure threshold

↓

Compare real data to prediction to check accuracy

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Physics-based modeling (2/3)

Unwanted reactions, such as the formation of a solid electrolyte interface (SEI) layer, can lead to battery degradation. Initially, a new battery will have an SEI layer which protects the battery from degrading. Elevated temperatures may cause modifications to the SEI layer. In addition, long term cycling can also cause excessive SEI growth and electrolyte consumption. High battery impedance will decrease efficiency and limit usable capacity as voltage limits will be reached more quickly. Degradation of a battery cell is a highly complex physical process, and arguably the hardest aspect of a battery to model physically.

This is where the complexity of physical modeling starts increasing and makes physical modeling of cell behavior difficult for modelers. Many of these physical cell processes can be expressed mathematically but are typically non-linear and coupled with one another. It therefore takes huge computational power to compute these processes.

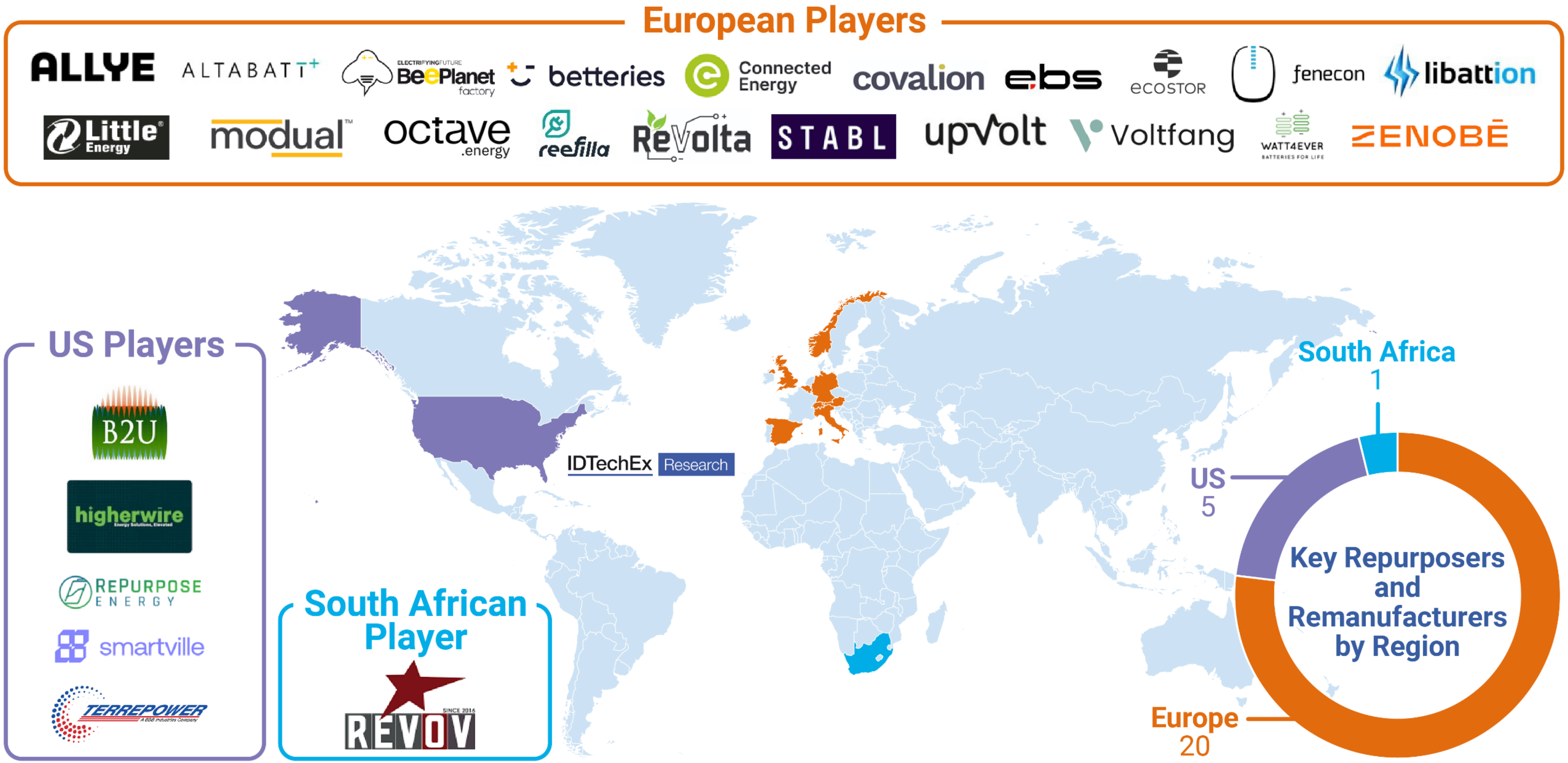
Challenges in Modeling Degradation

There are several unknowns and parameters involved:

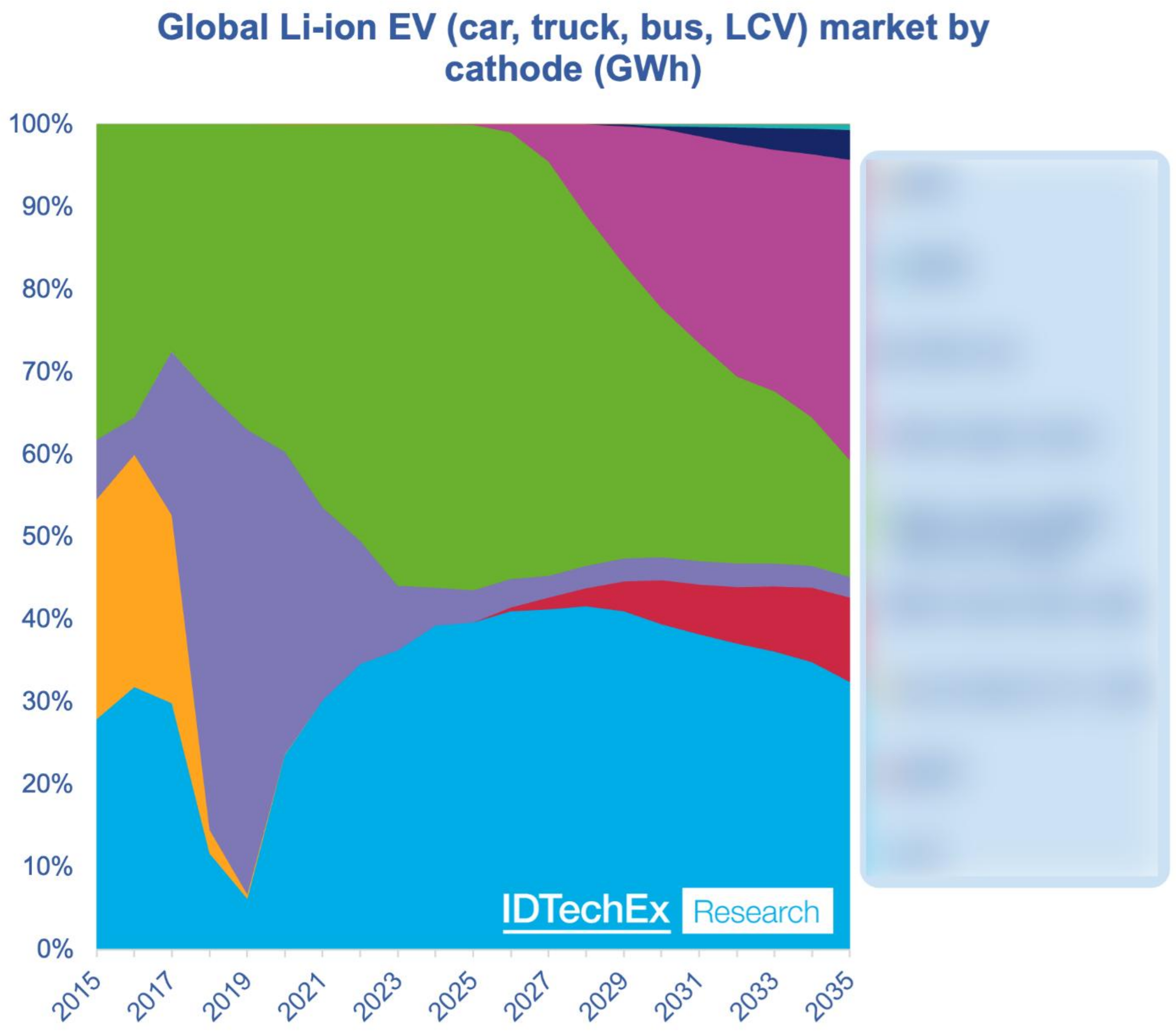
- SEI layer growth
- Li plating
- Loss of active material
- Surface cracking

These are not the only four processes which accelerate degradation but are arguably the four most important. All these processes are significantly impacted by the operating temperature of the battery, and deviations of operating temperature outside of battery specifications impact these processes more. Temperature is not the only factor that impacts degradation processes, and other factors such as the number of cycles completed, charge/discharge voltage/current and DOD also impact the rate of degradation.

Second-life repurposers and remanufacturers by HQ



Cathode market share in Li-ion for EVs



NMC and NCA have historically made up the majority of demand for Li-ion batteries in electric vehicles due to the superior energy density.

'Mid-nickel' NMC 532 and 622 will continue to be phased out in favor of higher nickel variants in BEV cars and for other EV segments.

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Second-life battery testing and assessment player summary

Included below is a summary of the players discussed in this chapter. It should be noted that many other players involved in cell-testing and battery management system diagnostics could also have capabilities for second-life assessment, as the techniques used are generally similar. The players included in the table below are those with the largest focuses in second-life assessment.

Player	Technology	Location	No. employees	Speed / accuracy	Notes

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Player overviews: HQ, founded date, total funding (US$M), employees, partnerships, projects, targets (1)

Player	HQ	Founded	Total Funding (US$M)	Employees	Other Key Details (Partnerships, Projects, Targets)
Allye Energy	UK	2021	[illegible]	[illegible]	[illegible]
Altabatt	Spain	2020	[illegible]	[illegible]	[illegible]
B2U Storage Solutions	US (CA)	2019	[illegible]	[illegible]	[illegible]
BBB Industries / TERREPOWER	US (AL)	1987	[illegible]	[illegible]	[illegible]
BeePlanet Factory	Spain	2018	[illegible]	[illegible]	[illegible]
Betteries	Germany	2018	[illegible]	[illegible]	[illegible]
Connected Energy	UK	2010	[illegible]	[illegible]	[illegible]
Covalion	Germany	2019	[illegible]	[illegible]	[illegible]

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Key automotive OEM activity (1/4): Audi, BMW, Ford, Honda, Hyundai

Automotive OEM	Key activities		
Audi	2023 pilot project; being co-run by company's non-profit German-Indian start-up Nunam, to use modules from Audi e-tron to electrify rickshaws in India. This is an example of a lower power electromobility application using second-life batteries.	2021; collaborated with RWE, Germany multinational energy company, to launch a pilot storage project of 4.5 MWh in Herdecke, Germany.	2019; 1.9 MWh second-life BESS installed in collaboration with The Mobility House.
BMW	[illegible]	[illegible]	
Ford	[illegible]		
Honda	[illegible]	[illegible]	
Hyundai	[illegible]	[illegible]	

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Second-life battery storage projects deployed by type of player

Second-life battery projects have been developed by repurposers, as well as other players including automotive OEMs, energy utilities, and V2G companies. This is important to highlight, as the graph shown right highlights a shift in the type of player responsible for developing second-life battery projects.

Before 2023, more projects were led by players whose focus is

MWh Second-Life Battery Projects Timeline by Repurposer vs 'One-Off' Projects from Other Players

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